

Question number	Answer	Mark
5(a)	The only correct answer is C lambda phage (λ phage) <i>A is incorrect because Ebola virus infects humans</i> <i>B is incorrect because the HIV infects humans</i> <i>D is incorrect because TMV infects plants</i>	(1)

Question number	Answer	Additional guidance	Mark
5(b)(i)	A description that includes two of the following points: - provide a {polar / hydrophilic} channel (1) - so that lysins can pass through the {non-polar / hydrophobic} {membrane / phospholipids / fatty acid tails} (out of cell) (1) - down their concentration gradient (1)	IGNORE direction of movement with respect to the cell	(2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An explanation that includes three of the following points: • primary structure is the sequence of amino acids that will determine the (tertiary) structure of {holin / protein} (1) • as this will determine the {bonds / position of bonds} (1) • (amino acids with) polar R groups will face into the channel (1) • (amino acids with) non-polar R groups will face outwards to the {fatty acids / phospholipids / membrane} (1)	PIECE TOGETHER DO NOT ACCEPT bases ACCEPT correctly named bond	(3)

Question number	Answer	Additional guidance	Mark
5(b)(iii)	An explanation that includes the following points: - lysins break bonds between the {peptidoglycan / murein} molecules (1) - therefore the virus particles {leave the bacterial cells / get (out) through the cell wall} (once formed) (1)	ACCEPT are enzymes that breakdown {peptidoglycan / murein} ACCEPT causing {bacterial cells to burst / pores in the cell wall}	